

# NB, NBG, TP, TPD

ATEX-approved pumps

Installation and operating instructions



Other languages

<http://net.grundfos.com/qr/i/96528411>

be  
think  
innovate

**GRUNDFOS** 

# English (GB) Installation and operating instructions

## Original installation and operating instructions

These installation and operating instructions describe ATEX-approved Grundfos NB, NBG, TP, TPD pumps and NB, NBG bare-shaft pumps.

Sections 1-5 provide important information about the product, information necessary to be able to unpack, install and start up the product in a safe way.

Sections 6-8 provide important information on service, fault finding and disposal of the product.

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Read this document before you install the product. Installation and operation must comply with local regulations and accepted codes of good practice.

## 1. General information

These supplementary installation and operating instructions apply to ATEX-approved Grundfos NB, NBG, TP, TPD pumps and NB, NBG bare-shaft pumps. The pumps comply with ATEX Directive 2014/34/EU.

### 1.1 Symbols used in this document

#### DANGER



Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.

#### WARNING



Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

#### CAUTION



Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The text accompanying the three hazard symbols DANGER, WARNING and CAUTION is structured in the following way:

### SIGNAL WORD



#### Description of hazard

Consequence of ignoring the warning.  
- Action to avoid the hazard.



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.

## 1.2 Related installation and operating instructions

In addition to these instructions, observe all installation and operating instructions supplied with the pump.

- NB, NBG  
product number: 96483177.
- TP, TPD  
product number: 96404999.
- TP  
product number: 96511031.

## 2. Product introduction

### 2.1 NB, NBG bare-shaft pumps

ATEX-approved NB, NBG bare-shaft pumps are supplied with an ATEX marking similar to that of the ATEX-approved NB, NBG pump. See section [2.5 Identification](#).

Installation and operating instructions mentioned in section [1.2 Related installation and operating instructions](#) also apply to ATEX-approved NB, NBG bare-shaft pumps.

### 2.2 Intended use

The pumps are suitable for use in areas or zones classified according to Directive 2014/34/EU. In case of doubt, consult the above-mentioned directives, or contact Grundfos.

The pumps must only be operated within the specification given in the "key application data sheet".

### 2.3 Pumped liquids

The pumps are suitable for thin, clean liquids, not containing solid particles or fibres.

### 2.4 Explosion protection documentation

The combination of NB, NBG, TP, TPD pumps and all monitoring equipment must be described in the explosion protection document according to the Directive 2014/34/EU. The responsibility rests with the installer or owner.

2.5 Identification

2.5.1 Nameplate

The nameplate on the pump head gives the following details:

- data for the standard pump
- data for the ATEX marking, pos. 1 and 2.

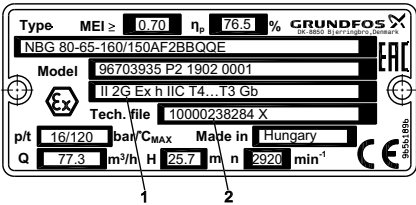


Fig. 1 Nameplate of ATEX-approved NBG pump

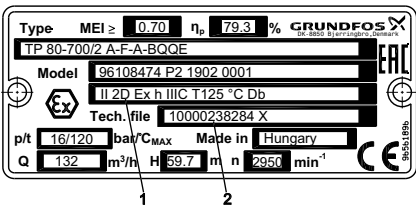


Fig. 2 Nameplate of ATEX-approved TP pump

Data for the ATEX marking only refer to the pump part including the coupling. The motor has a separate nameplate.

ATEX-related positions on the pump nameplate:

| Pos.        | Description                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1           | ATEX marking                                                                                                                   |
| II          | Equipment group                                                                                                                |
| 2, 3        | Equipment category                                                                                                             |
| G           | Environment:<br>Gas or vapours                                                                                                 |
| D           | Combustible dust                                                                                                               |
| Ex          | Explosion protection                                                                                                           |
| h           | Type of protection                                                                                                             |
| II C        | Environment group:<br>Gas or vapours                                                                                           |
| IIIC        | Combustible dust                                                                                                               |
| T4...T3     | Maximum surface temperature according to 80079-36. Temperature range or specific temperature.                                  |
| T125 °C     |                                                                                                                                |
| Gb          | EPL<br>(Equipment Protection Level)                                                                                            |
| Db          |                                                                                                                                |
| 2           | Technical file number                                                                                                          |
| 10000238284 | Number of technical file stored at DEKRA.                                                                                      |
| X           | "X" indicates that the equipment is subject to special conditions for safe use. The conditions are mentioned in this document. |

## 2.6 ATEX approvals

### 2.6.1 Scope of ATEX categories for NB, NBG, TP, TPD pumps

| Directive  |                                  | ATEX-approved NB, NBG, TP, TPD pumps |    |    |    |                                                                            |                            |                       |                            |
|------------|----------------------------------|--------------------------------------|----|----|----|----------------------------------------------------------------------------|----------------------------|-----------------------|----------------------------|
| 2014/34/EU | Equipment group                  | I                                    |    | II |    |                                                                            |                            |                       |                            |
|            | Equipment category               | M                                    |    | 1  |    | 2                                                                          |                            | 3                     |                            |
|            | Environment                      | 1                                    | 2  | G  | D  | G                                                                          | D                          | G                     | D                          |
|            | EPL (Equipment Protection Level) | Ma                                   | Mb | Ga | Da | Gb                                                                         | Db                         | Gc                    | Dc                         |
| 1999/92/EC | Zone                             |                                      |    | 0  | 20 | 1                                                                          | 21                         | 2                     | 22                         |
| Pumps      |                                  | None                                 |    |    |    | NB, NBG, TP, TPD                                                           |                            |                       |                            |
| Motors     |                                  | None                                 |    |    |    | II 2G Ex eb IIC T3 Gb<br>II 2G Ex db IIC T4 Gb<br>II 2G Ex db eb IIC T4 Gb | II 2D Ex tb IIC T125 °C Db | II 3G Ex ec IIC T3 Gc | II 3D Ex tc IIC T125 °C Dc |

The link between groups, categories and zones is explained in the 2014/34/EU Directive. Please note that this is a minimum directive. Some EU countries might therefore have stricter local rules. The user or installer is always responsible for checking that the group and category of the pump correspond to the zone classification of the installation site.

### 3. Installation requirements

#### 3.1 Location

##### 3.1.1 Horizontal or vertical installation

NB, NBG

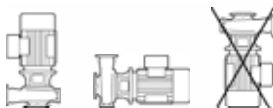


Fig. 3 Installation of NB, NBG pumps

TP, TPD

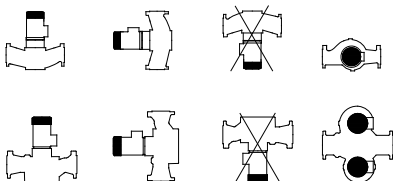


Fig. 4 Installation of a TP and a TPD pump with a motor size smaller than 11 kW

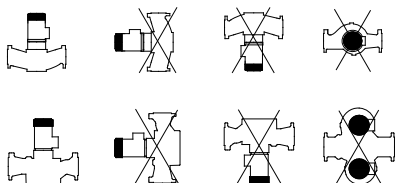


Fig. 5 Installation of a TP and a TPD pump with a motor size of 11 kW and above

##### 3.1.2 Pit installation

#### WARNING

##### Accumulation of explosive gases from shaft seal leakage



- Death or serious personal injury
- Provide adequate ventilation around the pump if the pump is installed in a pit. A minimum air exchange of 1.5 times per hour is required.

### 3.2 Bypass with pressure relief valve

#### CAUTION

##### Overheating



- Minor or moderate personal injury
- The pump must not run against a closed outlet valve or a closed shut-off element as this may cause overheating. Install a bypass with a pressure relief valve.

Observe the minimum flow rate. See section [1.2 Related installation and operating instructions](#).

### 4. Electrical connection

#### 4.1 Earthing the pump housing

#### DANGER

##### Electric shock



- Death or serious personal injury
- The pump housing must be earthed.

#### DANGER

##### Ignition of explosive environment



- Death or serious personal injury
- The pump housing must be earthed.



- Remove coating from the earthing point to ensure proper grounding connection.

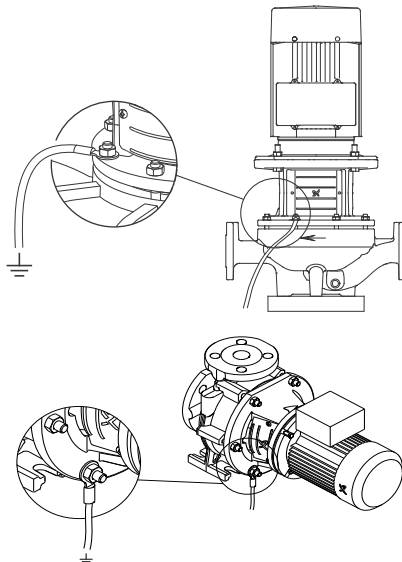


Fig. 6 Earthing point of the pump housing

Torque:  $80 \pm 16$  Nm.

TM07 4868 2519

TM00 3734 0897

TM00 3735 0897

TM05 4162 2112 - TM05 4161 2112

## 5. Starting up the product

### WARNING



#### Dry running

Death or serious personal injury

- Make sure that the pump is filled with pumped liquid during operation.

### 5.1 Before starting an ATEX-approved pump



Strictly observe the following check list.

1. Compare the order with the supplied pump and motor. Check that the EPL of the pump and motor (Equipment Protection Level) corresponds to what is ordered. If the EPL of the motor and pump differs from each other or the temperature class of the motor differs from that of the pump, the following applies:
  - The EPL which defines the lower protection level applies. Example: The EPL of the motor is Gc and the pump's is Gb. Gc applies.
  - The temperature class which defines the higher temperature applies.  
Example 1: Motor temperature class is T4 (135 °C), and pump temperature class is T3 (200 °C). T3 (200 °C) applies.  
Example 2: Motor temperature class is T3 (200 °C), and pump temperature class is T4...T3. T3 (200 °C) applies.
2. Check that the pumped liquid and its operating temperatures are in accordance with what is stated on the "key application data sheet".
3. Check that the shaft seal and rubber parts of the pump are as ordered. See the nameplate.

NB 32-125.1/142.1AEFAESBQOE

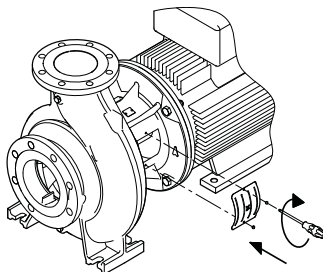
**Fig. 7** Example of code for rubber parts on the NB, NBG nameplate

TP 150-360/4 A-F-B-BQOE

**Fig. 8** Example of code for rubber parts on the TP nameplate

The key to the nameplate is shown in the installation and operating instructions for the standard pump.

4. Check that the maximum speed on the pump nameplate corresponds to the speed of the motor and that the pump will not be used for operation with a frequency over 60 Hz.
5. Check that the pump has been filled with liquid and is vented.



**Fig. 9** Location of vent screws

6. Check that the shaft can rotate freely. There must be no mechanical contact between impeller and pump housing.
7. Check the direction of rotation. The correct direction of rotation is shown by an arrow on the pump housing.
8. Before start and during operation, make sure that the pump does not leak or have any malfunctions.
9. The pump must be re-vented in these cases:
  - The pump has been stopped for a period of time.
  - Air or gas has accumulated in the pump.

TM03 3919 1206

TM07 4826 2519

TM06 7168 3116

## 5.2 Non-flammable liquids

### EPL Gb/Db (category 2G/D)

If the operator cannot ensure that the pump is filled with pumped liquid during operation, appropriate monitoring, for instance dry-running protection, is required to stop the pump in case of malfunction.

### EPL Gc/Dc (category 3G/D)

No additional monitoring is required.

## 5.3 Flammable liquids

### EPL Gb/Db, Gc/Dc

If the operator cannot ensure that the pump is filled with pumped liquid during operation, appropriate monitoring, for instance dry-running protection, is required to stop the pump in case of malfunction. Ventilation around the pump is also required. The leakage rate of a normally working shaft seal is less than 36 ml for each 24 hours of operation. A minimum air exchange of 1.5 times per hour is required.

### CAUTION

#### Flammable material

Minor or moderate personal injury

- The responsibility for checking the functions of the dry-running protection, such as flow rate, sealing pressure and temperature of the barrier or flushing liquid rests with the installer or owner.



Increased leakage can be the indication of shaft seal failure. For some types of liquids the leakage will not be visible due to evaporation.

## 5.4 Checking the direction of rotation



Never check the direction of rotation by starting the pump, not even for a short period, unless the pump has been filled with liquid. This is to prevent temperature rises resulting from contact between rotating and stationary components, and to protect the shaft seal against dry running.

## 6. Servicing the product

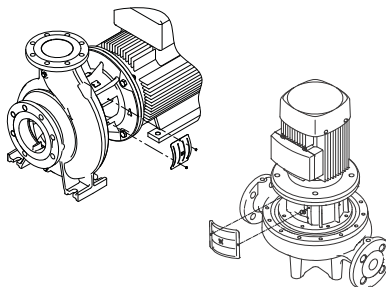
Service on the pump end can be made on site. The pump end does not need to be shipped to an approved ATEX workshop.



Strictly observe the following check list.

It may be overruled by stricter local maintenance schedules.

1. Check on a daily basis that the shaft seal functions correctly.
2. Any standby pump installed must be switched on once a week to keep it operational.
3. It is the responsibility of the customer to do the following:
  - Decide whether to use the non-sparking tools or to shut down the system for service.
  - Lay down a cleaning scheme for pump surfaces in installations with combustible dust.
4. When cleaning a pump in a combustible dust environment, remember to take the shaft guard off and clean the shaft cavity.



**Fig. 10** Take the shaft guard off

5. Inspect O-rings for cracks, elasticity, and permanent change of shape when doing periodic maintenance or servicing the pump. Renew O-rings if necessary.
6. Torques for all fasteners can be found in the service instructions for NB, NBG and TP, TPD products.

TM07 4867 2519 - TM07 4869 2519



## 7. Technical data

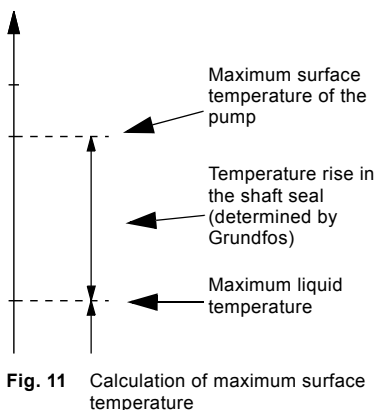
### 7.1 Operating conditions

#### 7.1.1 Liquid temperature

The maximum liquid temperature depends on the temperature class specified by customer and the shaft seal.

| Temperature class | Maximum surface temperature [°C] |
|-------------------|----------------------------------|
| T1                | 450                              |
| T2                | 300                              |
| T3                | 200                              |
| T4                | 135                              |
| T5                | 100                              |
| T6                | 85                               |

The illustration below shows the maximum surface temperature of the pump as a result of maximum liquid temperature and temperature rise in the shaft seal.



**Fig. 11** Calculation of maximum surface temperature

The maximum surface temperature of the pump must be at least 5 °C lower than the maximum surface temperature of the temperature class specified by the customer.

The maximum liquid temperature and the temperature class specified by the customer are stated on the "key application data sheet" supplied with the pump. See the example at the end of this document.

A copy is filed by Grundfos and can be traced by means of the product number and serial number on the pump nameplate.

Model B 96689648 P2 07 02 0001

**Fig. 12** Model, product number, production site, year, week, and serial number



Do not exceed the maximum liquid temperature or pump another type of liquid than those specified on the "key application data sheet" supplied with the pump. Damage resulting from disregarding this warning will not be covered by the Grundfos warranty.



If the "key application data sheet" is missing, contact Grundfos for information about the maximum liquid temperature.

If the pump is to be operated at a higher liquid temperature, or with another liquid than the one stated on the data sheet, contact Grundfos.

#### 7.1.2 Ambient temperature

The ambient temperature range in operation is -20 to +60 °C for the pump-end.

## 8. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.



The crossed-out wheeled bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.

TM04 0062 4907

## Appendix

## Example of key application data sheet

Dear customer, please fill in the following questionnaire in cooperation with a Grundfos representative. This will help to ensure that Grundfos supplies you with a pump solution adapted to meet exactly your needs in terms of pump type, pump materials, shaft seal arrangement, shaft seal type, elastomers and accessories.

### Customer information

|                  |                   |
|------------------|-------------------|
| Company name:    | Project title:    |
| Customer number: | Reference number: |
| Phone number:    | Customer contact: |
| Fax number:      |                   |
| E-mail address:  |                   |

### Quotation made by:

|                 |                   |           |
|-----------------|-------------------|-----------|
| Company name:   | Prepared by:      |           |
| Phone number:   | Date:             | Page 1 of |
| Fax number:     | Quotation number: |           |
| E-mail address: |                   |           |

## Operating conditions

### Pumped liquid

|                                                   |                                                                                                    |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Type of liquid:                                   | <b>Water</b>                                                                                       |
| Chemical composition (if available):              |                                                                                                    |
| Distilled/demineralised water                     | Yes ____ No ____                                                                                   |
| Conductivity of distilled/<br>demineralised water | _____ [μS/cm]                                                                                      |
| Minimum liquid temperature:                       | _____ [°C]                                                                                         |
| Maximum liquid temperature:                       | <b>110</b> _____ [°C]                                                                              |
| Vapour pressure of liquid:                        | _____ [bar]                                                                                        |
| Liquid concentration:                             | _____ %                                                                                            |
| Liquid pH value:                                  | _____                                                                                              |
| Liquid viscosity:                                 | Dynamic viscosity: _____ [cP] = [mPa·s]<br>Kinematic viscosity: _____ [cSt] = [mm <sup>2</sup> /s] |
| Liquid density:                                   | _____ [kg/m <sup>3</sup> ]                                                                         |
| Specific heat capacity of liquid:                 | _____ [kJ/(kg·K)]                                                                                  |
| Air/gas in liquid?                                | Yes <b>X</b> No ____                                                                               |
| Solids in liquid?                                 | Yes ____ No <b>X</b>                                                                               |
| Contents of solids in liquid (if available):      | _____ % of mass                                                                                    |
| Additives in liquid?                              | Yes ____ No <b>X</b>                                                                               |
| Does the liquid crystallise?                      | Yes ____ No <b>X</b>                                                                               |
| When does crystallisation happen?                 | _____<br>_____                                                                                     |

Does the liquid get sticky when volatiles evaporate from the pumped liquid?

Yes \_\_\_\_\_ No \_\_\_\_\_

Description of 'sticky' circumstances:

\_\_\_\_\_  
\_\_\_\_\_

Is the liquid hazardous/poisonous?

Yes \_\_\_\_\_ No \_\_\_\_\_

Special measures to be taken into account when dealing with this hazardous/poisonous liquid:

\_\_\_\_\_  
**Pay attention to that air may build up - especially at stand stills, and make appropriate monitoring to avoid dry running.**

Special measures for handling this liquid:

\_\_\_\_\_

### CIP liquid (cleaning in place)

Type of liquid:

\_\_\_\_\_

Chemical composition (if available):

\_\_\_\_\_

Liquid Temperature during operation:

\_\_\_\_\_ [°C]

Maximum liquid temperature:

\_\_\_\_\_ [°C]

Vapour pressure of liquid:

\_\_\_\_\_ [bar]

Liquid concentration:

\_\_\_\_\_ %

Liquid pH value:

\_\_\_\_\_

### Pump sizing

Main duty point

Q: \_\_\_\_\_ [m<sup>3</sup>/h] H: \_\_\_\_\_ [m]

Max. duty point

Q: \_\_\_\_\_ [m<sup>3</sup>/h] H: \_\_\_\_\_ [m]

Min. duty point

Q: \_\_\_\_\_ [m<sup>3</sup>/h] H: \_\_\_\_\_ [m]

### Ambient operating conditions

Ambient temperature:

\_\_\_\_\_ [°C]

Altitude above sea level:

\_\_\_\_\_ [m]

### Pressure

Minimum inlet pressure:

\_\_\_\_\_ [bar]

Maximum inlet pressure:

\_\_\_\_\_ [bar]

Discharge pressure (inlet pressure + head):

\_\_\_\_\_ [bar]

### ATEX marking

#### Required marking of the pump

Customer's equipment group (e.g.: II):

\_\_\_\_\_ **II** \_\_\_\_\_

Customer's equipment category (e.g.: 2, 3)

\_\_\_\_\_ **2** \_\_\_\_\_

Gas (G) and/or dust (D)

Gas (G) **X** Dust (D) \_\_\_\_\_ Gas and dust (G/D) \_\_\_\_\_

**Required marking of the motor**

Protection type (e.g.: d, de, e, nA) \_\_\_\_\_ **e** \_\_\_\_\_

Maximum experimental safe gap (e.g.: B, C) \_\_\_\_\_

Temperature class - gas (e.g.: T3, T4, T5) \_\_\_\_\_ **T4** \_\_\_\_\_

Temperature class - dust (e.g.: 125 °C) \_\_\_\_\_ [°C] \_\_\_\_\_

**Description/sketch**

Detailed description of ATEX application (attach a drawing if possible) \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**ATEX certificate required**

Yes **X** No \_\_\_\_\_

**Frequency converter**

Frequency converter option wanted? Yes \_\_\_\_\_ No **X** \_\_\_\_\_

Control parameter: Pressure \_\_\_\_\_ Temperature \_\_\_\_\_ Flow \_\_\_\_\_ Other \_\_\_\_\_

Detailed description of requirements: (attach a drawing if possible) \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**System information**

Please provide us with some information about your system and maybe a simple sketch. This will give us hints as to whether you need accessories or monitoring equipment, or whether you already have a suitable system which makes it unnecessary to attach any further equipment.



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